

Bidding Metrics

2024-06-12

- Let n_i^{task} = Number of projects of the same task held by the contractor in the three quarters before starting project i
- Let n_i = Number of projects held by the contractor in the three quarters before starting project i
- Define:
 - Inexperienced contractor $_i$ = 1 if $n_i^{task} = 0$
 - New entrant $_i$ = 1 if $n_i = 0$
 - Level of inexperience $_i = 1 - \frac{n_i^{task}}{n_i}$ if $n_i > 0$
 - Level of inexperience $_i = 1$ if $n_{i,k,t} = 0$.

1 Effect on Bids

	Num offers	Initial Duration	Initial Budget (000s)
	I	II	III
Treat	0.65*** (0.19)	-21.39*** (1.14)	-152.37*** (10.05)
New Entrant	-0.25 (0.49)	3.34 (3.44)	182.84*** (34.16)
Treat $\times$ SA $_i$	0.55** (0.24)	1.65 (1.52)	-35.89** (14.30)
Treat $\times$ New Entrant	0.39 (0.50)	-5.18 (3.76)	-103.26*** (36.19)
SA $_i$ $\times$ New Entrant	1.71*** (0.59)	5.97 (5.84)	141.62** (60.65)
Treat $\times$ SA $_i$ $\times$ New Entrant	-1.75*** (0.63)	-12.42* (6.42)	-81.16 (64.44)
Task FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	142,660	140,787	144,025
R ²	0.34	0.26	0.25
Within R ²	0.00	0.01	0.01

Clustered (Project ID) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

	Num offers	Initial Duration	Initial Budget (000s)
	I	II	III
Treat	0.97*** (0.23)	-23.93*** (1.22)	-128.67*** (10.42)
Inexperienced contractor	1.34*** (0.38)	-4.63** (2.32)	218.71*** (23.01)
Treat $\times$ SA _i	0.37 (0.26)	4.42*** (1.58)	-47.78*** (14.64)
Treat $\times$ Inexperienced contractor	-0.84** (0.38)	6.16** (2.51)	-115.85*** (23.94)
SA _i $\times$ Inexperienced contractor	0.23 (0.54)	7.82** (3.76)	19.47 (38.36)
Treat $\times$ SA _i $\times$ Inexperienced contractor	-0.28 (0.56)	-15.62*** (4.12)	1.31 (40.65)
Task FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	142,660	140,787	144,025
R ²	0.34	0.26	0.25
Within R ²	0.00	0.01	0.01

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

2 Effect on Delays

2.1 Old vs New Entrant

	Percentage delay rate							
	Old entrants				New entrants			
	I	II	III	IV	V	VI	VII	VIII
Constant	9.904*** (0.142)				10.895*** (0.296)			
Treat	-2.242*** (0.132)	-2.251*** (0.132)	-0.386*** (0.138)	-0.414*** (0.138)	-0.531* (0.273)	-0.566** (0.271)	-0.739** (0.304)	-0.791** (0.307)
SA _i	0.702*** (0.176)	1.211*** (0.190)	1.209*** (0.189)	1.158*** (0.188)	2.671*** (0.497)	3.857*** (0.519)	3.905*** (0.558)	3.845*** (0.557)
Ln(Stage)	2.824*** (0.035)	2.913*** (0.038)	2.757*** (0.036)	2.728*** (0.036)	3.778*** (0.085)	4.012*** (0.092)	3.873*** (0.094)	3.849*** (0.094)
Post	-1.234*** (0.173)				-0.930** (0.369)			
Treat $\times$ Post	0.439** (0.197)	0.446** (0.197)	0.180 (0.190)	0.170 (0.190)	-0.605 (0.420)	-0.562 (0.420)	0.122 (0.432)	0.105 (0.432)
Treat $\times$ SA _i $\times$ Post	0.725*** (0.202)	0.718*** (0.202)	0.804*** (0.199)	0.813*** (0.198)	0.238 (0.562)	0.172 (0.561)	-0.236 (0.595)	-0.208 (0.596)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	154,113	154,113	154,113	154,113	30,769	30,769	30,769	30,769
R ²	0.06	0.06	0.13	0.13	0.07	0.07	0.14	0.14
Within R ²		0.05	0.05	0.05		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

3 Experienced vs Inexperienced

	Percentage delay rate							
	Experienced contractors				Inexperienced contractors			
	I	II	III	IV	V	VI	VII	VIII
Constant	9.657*** (0.172)				10.697*** (0.193)			
Treat	-2.433*** (0.164)	-2.424*** (0.164)	-0.033 (0.169)	-0.094 (0.169)	-1.370*** (0.173)	-1.401*** (0.172)	-0.906*** (0.187)	-0.893*** (0.188)
SA _i	0.500** (0.199)	0.830*** (0.217)	0.518** (0.211)	0.484** (0.210)	2.045*** (0.312)	3.201*** (0.327)	3.363*** (0.346)	3.289*** (0.347)
Ln(Stage)	2.657*** (0.041)	2.712*** (0.043)	2.494*** (0.041)	2.466*** (0.041)	3.487*** (0.055)	3.699*** (0.059)	3.675*** (0.060)	3.657*** (0.059)
Post	-1.170*** (0.209)				-1.206*** (0.240)			
Treat × Post	0.371 (0.236)	0.352 (0.237)	-0.016 (0.226)	-0.027 (0.225)	0.220 (0.273)	0.257 (0.273)	0.412 (0.277)	0.433 (0.277)
Treat × SA _i × Post	0.909*** (0.227)	0.917*** (0.227)	0.874*** (0.220)	0.888*** (0.219)	0.479 (0.359)	0.373 (0.359)	0.124 (0.375)	0.179 (0.377)
Time FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Task FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	113,016	113,016	113,016	113,016	71,866	71,866	71,866	71,866
R ²	0.05	0.05	0.13	0.14	0.07	0.07	0.12	0.13
Within R ²		0.05	0.04	0.04		0.07	0.06	0.06

Clustered (Project ID) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1